Srijan Challapalli

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EDUCATION

Purdue University **West Lafayette, IN**
B.S. in Artificial Intelligence *Aug. 2025 - May 2028 (Expected)*
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Linear Algebra, Probability & Statistics, Discrete Math

SKILLS

Languages: Python, C++, Java, SQL, TypeScript/JavaScript, Rust, Swift
Frameworks: React, Next.js, FastAPI, Flask, Express, SwiftUI
Data & ML: Pandas, NumPy, XGBoost, PostgreSQL, Supabase, LlamaParse
Tools & Testing: Git/GitHub, Linux, n8n, Vercel, Vitest, Criterion, unit and integration testing

EXPERIENCE

MergeWorks **Remote**
Software Engineer Intern *May 2026 - Present*
- Built an M&A due-diligence pipeline that transforms financial packets, including Excel files, QuickBooks exports, tax returns, and scanned PDFs, into source-cited analysis in 71 seconds at p50 and 125 seconds at p95, replacing hours of manual review.
- Improved report accuracy by building automated checks that verify every financial figure against its source documents (five reconciliation checks, 2% tolerance), making each reported number traceable and audit-ready.
- Led a zero-downtime migration from n8n Data Tables to PostgreSQL and Supabase, retired six legacy workflows, prevented duplicate paid AI calls, and maintained quality control with 111 passing unit tests.

Cosmos Granite & Marble **Chantilly, VA**
Software Engineer Intern *May 2025 - Aug. 2025*
- Built a Python data-collection pipeline using SerpAPI and web scraping to identify 1,000+ prospective customers across target states, generating qualified leads that supported an 18% increase in sales.
- Automated extraction, filtering, and deduplication of business contact data into state-level Excel datasets that powered targeted outreach campaigns, contributing to a 15% increase in click-through rates.
- Partnered with the sales team across 50+ client consultations to gather feedback and refine the pipeline's lead-targeting and filtering criteria.

PROJECTS

ChitYap: *Swift, SwiftUI, Supabase, PostgreSQL, Apple Vision, HealthKit, APNs*
- Co-founded and shipped a privacy-first social accountability iOS app used by 300+ beta users to create nearly 5,000 posts; built real-time feeds, friendships, notifications, and offline retry using SwiftUI, Supabase, and PostgreSQL.
- Built an on-device Apple Vision rep counter running at 15-30 FPS and optimized 60+ row-level security policies by reducing authentication checks to one evaluation per query.
- Designed a modular Views, Managers, Services, and API architecture backed by approximately 180 SQL migrations, 17 feed indexes, and automated architecture tests.

Prediction Market: *Solidity, TypeScript, Next.js, PostgreSQL, Hardhat*
- Built an Ethereum testnet prediction market supporting 8 concurrent user-created markets with ERC20 outcome shares, collateral-backed trading, closing deadlines, settlement, and winner redemption.
- Validated collateral coverage and single-use redemption across 7,500 simulated trades, including rejected transactions and invalid-market refunds.
- Built a PostgreSQL event indexer with 1.8-second p95 latency from event detection to persistence, including duplicate-event handling, restart recovery, and portfolio tracking.

LSM-Tree Storage Engine: *Rust, C++, Tokio, Criterion, CMake, Linux perf*
- Building a persistent key-value store in Rust with a write-ahead log, in-memory memtable, immutable SSTables, Bloom filters, background compaction, and crash recovery.
- Benchmarking throughput and p99 latency with Criterion and Linux perf while using WAL replay to preserve acknowledged writes after simulated crashes.
- Ported the core storage path to C++17 on Linux - memtable, SSTable reader/writer, and Bloom filter - building with CMake and validating byte-compatible SSTable output against the Rust implementation.

ADDITIONAL INFORMATION

Former No. 1 nationally ranked badminton player; two-time national semifinalist. Interests include weightlifting, hiking, and football.